

Planetary Gearbox - 4-Stage Assembly Checklist

A. Gearbox information

Customer name

Work order number

Manufacturer

Model

Serial number

Reduction ratio (overall)

 : 1

Stage 1 ratio

 : 1

Stage 2 ratio

 : 1

Stage 3 ratio

 : 1

Stage 4 ratio

 : 1

Oil capacity (gal / L)

Oil grade

1. Prep and inspection

- ☐ Work order, drawings, BOM, and assembly spec on hand at current revision
- ☐ Clean covered work area, lifting gear rated and inspected
- ☐ Housing cleaned and deburred, oil passages, drain and breather ports clear
- ☐ Parts kept by planetary stage, match marks verified
- ☐ Sun, planet and ring gear teeth and splines inspected, no damage or burrs
- ☐ Planet pins, pin bores and needle rollers checked; rollers counted per planet
- ☐ Thrust washers checked for wear and thickness
- ☐ Bearings checked for part number, cups and cones kept as matched sets
- ☐ Measuring tools calibrated (micrometers, bore gauges, dial indicators, torque wrench)

2. Input shaft and seal

☐ Input shaft bearings installed and fully seated

☐ Input shaft endplay set:

Endplay:

☐ Input seal installed (lip direction correct, lubricated, no damage)

☐ Input shaft rotates freely

P1. Planetary stage 1

Sun gear teeth (Z sun)

Ring gear teeth (Z ring)

Planet gear teeth

Number of planets

Stage ratio, calculated ($1 + Z \text{ ring} / Z \text{ sun}$)

: 1

Stage ratio, measured (input turns per carrier turn)

: 1

☐ Planet pins installed with correct fit and retained (roll pins / circlips / plates)

☐ Planet bearings / needle rollers installed, complete count, lubricated

☐ Thrust washers installed on both sides of each planet (thickness recorded below)

☐ Planets rotate freely on their pins, no binding

☐ Sun gear installed; sun float / axial play checked

☐ Ring gear installed: mounting fit, dowels fitted, bolts torqued:

Bolt torque:

☐ Planet carrier installed; carrier bearing seated

☐ Coupling / spline, stage 1 to stage 2 engaged, full spline engagement

Ram Gear Manufacturing Assembly Verification Data

Stage 1 measurements

Measurement	Measured	Spec	OK
Sun-planet backlash			☐
Planet-ring backlash			☐
Sun gear float / axial play			☐
Carrier endplay			☐
Thrust washer thickness			☐

Stage 1 planets (one row per planet)

Planet	Pin fit (measured)	Needle rollers (count / OK)	Thrust washer thickness	Rotates freely
Planet 1				☐
Planet 2				☐
Planet 3				☐
Planet 4				☐
Planet 5				☐
Planet 6				☐

P2. Planetary stage 2

Sun gear teeth (Z sun)	Ring gear teeth (Z ring)	Planet gear teeth	Number of planets

Stage ratio, calculated ($1 + Z \text{ ring} / Z \text{ sun}$)

: 1

Stage ratio, measured (input turns per carrier turn)

: 1

- ☐ Planet pins installed with correct fit and retained (roll pins / circlips / plates)
- ☐ Planet bearings / needle rollers installed, complete count, lubricated
- ☐ Thrust washers installed on both sides of each planet (thickness recorded below)

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- ☐ Planets rotate freely on their pins, no binding
- ☐ Sun gear installed; sun float / axial play checked
- ☐ Ring gear installed: mounting fit, dowels fitted, bolts torqued:

Bolt torque:

- ☐ Planet carrier installed; carrier bearing seated
- ☐ Coupling / spline, stage 2 to stage 3 engaged, full spline engagement

Stage 2 measurements

Measurement	Measured	Spec	OK
Sun-planet backlash			☐
Planet-ring backlash			☐
Sun gear float / axial play			☐
Carrier endplay			☐
Thrust washer thickness			☐

Stage 2 planets (one row per planet)

Planet	Pin fit (measured)	Needle rollers (count / OK)	Thrust washer thickness	Rotates freely
Planet 1				☐
Planet 2				☐
Planet 3				☐
Planet 4				☐
Planet 5				☐
Planet 6				☐

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P3. Planetary stage 3

Sun gear teeth (Z sun)

Ring gear teeth (Z ring)

Planet gear teeth

Number of planets

Stage ratio, calculated (1 + Z ring / Z sun)

 : 1

Stage ratio, measured (input turns per carrier turn)

 : 1

- ☐ Planet pins installed with correct fit and retained (roll pins / circlips / plates)
- ☐ Planet bearings / needle rollers installed, complete count, lubricated
- ☐ Thrust washers installed on both sides of each planet (thickness recorded below)
- ☐ Planets rotate freely on their pins, no binding
- ☐ Sun gear installed; sun float / axial play checked
- ☐ Ring gear installed: mounting fit, dowels fitted, bolts torqued:

Bolt torque:

- ☐ Planet carrier installed; carrier bearing seated
- ☐ Coupling / spline, stage 3 to stage 4 engaged, full spline engagement

Stage 3 measurements

Measurement	Measured	Spec	OK
Sun-planet backlash			☐
Planet-ring backlash			☐
Sun gear float / axial play			☐
Carrier endplay			☐
Thrust washer thickness			☐

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Stage 3 planets (one row per planet)

Planet	Pin fit (measured)	Needle rollers (count / OK)	Thrust washer thickness	Rotates freely
Planet 1				☐
Planet 2				☐
Planet 3				☐
Planet 4				☐
Planet 5				☐
Planet 6				☐

P4. Planetary stage 4

Sun gear teeth (Z sun)

Ring gear teeth (Z ring)

Planet gear teeth

Number of planets

Stage ratio, calculated ($1 + Z \text{ ring} / Z \text{ sun}$)

: 1

Stage ratio, measured (input turns per carrier turn)

: 1

- ☐ Planet pins installed with correct fit and retained (roll pins / circlips / plates)
- ☐ Planet bearings / needle rollers installed, complete count, lubricated
- ☐ Thrust washers installed on both sides of each planet (thickness recorded below)
- ☐ Planets rotate freely on their pins, no binding
- ☐ Sun gear installed; sun float / axial play checked
- ☐ Ring gear installed: mounting fit, dowels fitted, bolts torqued:

Bolt torque:

- ☐ Planet carrier installed; carrier bearing seated
- ☐ Output spline / coupling (stage 4 carrier to output) engaged, full spline engagement

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Stage 4 measurements

Measurement	Measured	Spec	OK
Sun-planet backlash			☐
Planet-ring backlash			☐
Sun gear float / axial play			☐
Carrier endplay			☐
Thrust washer thickness			☐

Stage 4 planets (one row per planet)

Planet	Pin fit (measured)	Needle rollers (count / OK)	Thrust washer thickness	Rotates freely
Planet 1				☐
Planet 2				☐
Planet 3				☐
Planet 4				☐
Planet 5				☐
Planet 6				☐

3. Output shaft, bearings and seals

☐ Output shaft bearings installed and fully seated

☐ Output shaft endplay / preload set:

Endplay / preload:

☐ Output seals installed (lip direction correct, lubricated, no damage)

☐ Output shaft turns smoothly with the input

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4. Bearing shims

Bearing shim record

Shaft	Location	Starting shim (in/mm)	Final shim (in/mm)	Measured endplay/preload	Spec	OK	Shim pack	
Input	Drive end					☐	☐	Replace ☐ Reuse ☐
Input	Non-drive end					☐	☐	Replace ☐ Reuse ☐
Output	Drive end					☐	☐	Replace ☐ Reuse ☐
Output	Non-drive end					☐	☐	Replace ☐ Reuse ☐
Stage 1 carrier	Carrier bearing					☐	☐	Replace ☐ Reuse ☐
Stage 2 carrier	Carrier bearing					☐	☐	Replace ☐ Reuse ☐
Stage 3 carrier	Carrier bearing					☐	☐	Replace ☐ Reuse ☐
Stage 4 carrier	Carrier bearing					☐	☐	Replace ☐ Reuse ☐

5. Housing and lubrication

☐ Housing closed with correct gaskets / sealant, bolts torqued:

Bolt torque:

☐ Breather, plugs and sight glass installed

☐ Filled with the specified oil:

Grade:

Quantity:

☐ No leaks after fill

6. Final checks and run test

☐ Rotates smoothly by hand through a full output revolution

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☐ Output rotational backlash measured (input locked):

Backlash:

☐ Run test: noise, vibration and temperature acceptable:

Temperature rise:

☐ Direction of rotation and ratio verified

☐ Serial numbers, shim thicknesses, endplay, backlash, torques and test data recorded

Notes

Notes

Sign-off

Assembler

Name

Signature (type name)

Date

Inspector

Name

Signature (type name)

Date